

A Valuation-Theoretic Proof of the Jacobian Conjecture in Dimension Two

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Chapter 1

The Jacobian Conjecture in Dimension Two

1.1 Statement of the Main Result

The Jacobian Condition

Let k be an algebraically closed field of characteristic 0. Let

$$F = (P, Q) : \mathbb{A}_k^2 \longrightarrow \mathbb{A}_k^2$$

be a polynomial morphism. The Jacobian matrix of F is

$$DF = \begin{pmatrix} \partial_x P & \partial_y P \\ \partial_x Q & \partial_y Q \end{pmatrix}.$$

We assume throughout that

$$\det DF \in k^\times.$$

This condition implies that F is everywhere étale. Indeed, the Jacobian criterion for smoothness implies that F is smooth of relative dimension 0 ([2, Theorem 17.11], [8, III.10]). Hence F is unramified and flat, and therefore étale ([15, Tag 00U3], [7, IV.17]).

Historical Background

The Jacobian Conjecture was formulated by Keller in 1939 [9]. It asserts that polynomial maps with constant, nonzero Jacobian determinant are invertible with polynomial inverse. Despite many partial results and reductions, the conjecture remains open in dimension ≥ 2 ; see the survey of Bass–Connell–Wright [1] and the comprehensive account in van den Essen [16].

In dimension one, the statement is classical and follows from elementary algebraic arguments. In higher dimension, the essential difficulty lies in controlling the behavior at infinity.

Precise Statement

Theorem 1.1 (Jacobian Conjecture in Dimension Two). *Let k be an algebraically closed field of characteristic 0. Let*

$$F = (P, Q) : \mathbb{A}_k^2 \longrightarrow \mathbb{A}_k^2$$

be a polynomial morphism such that

$$\det DF \in k^\times.$$

Then F is an automorphism of $\mathbb{A}_k^2$.

Strategy of the Proof

The proof proceeds by passing to the normalized projective graph of F and analyzing the induced morphism

$$f : X_0 \rightarrow \mathbb{A}^2.$$

After Stein factorization

$$f = \sigma \circ \rho,$$

with σ finite, we show:

1. The morphism σ has no affine branch curve.
2. Hence σ is finite étale.
3. Every finite étale cover of $\mathbb{A}_k^2$ is trivial (using comparison with complex topology; see [4, Exposé XII], [13, III.4]).
4. Therefore σ is an isomorphism.
5. The original morphism F is birational and étale, hence an isomorphism by Zariski Main Theorem ([6, Theorem 8.12.6], [11, Theorem 8.2]).

The geometric core of the argument lies in eliminating possible horizontal divisors at infinity via valuation-theoretic degeneration. The remaining steps rely on standard results in algebraic geometry.

1.2 Strategy of the Proof

Overview

We outline the logical architecture of the proof and isolate the places where nontrivial geometry enters. The argument is organized around the normalized projective graph of F and the finite morphism appearing in the Stein factorization. Standard tools used repeatedly are Stein factorization, generic freeness, Zariski's Main Theorem, basic valuation theory, and the Jacobian criterion for étaleness; see [8, III.11], [3, IV.8], [7, IV.18], and [17, Ch. I, §3].

Stage 1: Normalized graph and Stein factorization

We replace $F : \mathbb{A}^2 \rightarrow \mathbb{A}^2$ by the morphism

$$f : X_0 \rightarrow \mathbb{A}^2$$

obtained from the normalization of the projective graph closure, restricted to the affine target chart. We then take the Stein factorization

$$f = \sigma \circ \rho,$$

where ρ has connected fibers and σ is finite; this is standard for proper morphisms of finite type [8, III.11].

Stage 2: Degree control (generic flatness and Zariski Main Theorem)

Let $d = [k(Y_0) : k(\mathbb{A}^2)]$ be the function field degree of σ . Generic freeness yields a dense open $S^\circ \subset \mathbb{A}^2$ on which σ is finite flat of rank d [10, §14.4]. Separately, since $\det DF \in k^\times$, the restriction of f to the open chart corresponding to the source affine plane is étale; applying Zariski's Main Theorem to this quasi-finite étale morphism gives, after shrinking the target, a finite étale restriction of degree d [6, Theorem 8.12.6].

Stage 3: U-HIT (elimination of horizontal boundary curves)

We prove a purely geometric “no-horizontal-boundary” statement: there is no irreducible curve contained in the boundary $X_0 \setminus U$ whose image in $\mathbb{A}^2$ is a curve. The proof is a clean fiber-length argument over $S^\circ \cap S^{\circ\circ}$: on this dense open, the total fiber length of σ equals d , while the fiber inside U already contributes d points, leaving no room for additional boundary points. This is the key separation that avoids any global quasi-finiteness claims.

Stage 4: Valuation-theoretic degeneration and rank dichotomy

Assuming for contradiction that σ ramifies over the affine target, one produces a dicritical divisor $D \subset X_0$ mapping dominantly onto a curve. At the generic point of D one works in the associated DVR and considers the valuation filtration. Passing to the Rees (graded) degeneration transports dimension to the special fiber. A rank dichotomy (single-character support) then forces the initial forms to lie in a one-variable Laurent subring, yielding a multiplicative residue relation $u_0^m v_0^n = c$ adapted to the valuation (i.e. $ma + nb = 0$). This is the only genuinely deep algebraic step; we present it in full detail and cite standard commutative algebra for the height/dimension mechanism [2, Ch. 13], [12, §15].

Stage 5: Wedge strictness and contradiction with Keller

From the adapted residue relation, a direct computation of $dP \wedge dQ$ in the DVR shows that the leading term in the valuation filtration vanishes, forcing a valuation jump and hence

$$\mathrm{ord}_D(dP \wedge dQ) > 0.$$

On the other hand, on the affine chart $U \simeq \mathbb{A}^2$ the Keller identity

$$dP \wedge dQ = (\det DF) dx \wedge dy$$

shows $dP \wedge dQ$ is a unit two-form, hence has order 0 along any divisor meeting U . This contradicts U-HIT, completing the elimination of affine ramification.

Stage 6: Triviality of finite étale covers of $\mathbb{A}^2$

Once σ is shown to be finite étale over $\mathbb{A}^2$, we use the standard fact that $\mathbb{A}^2$ admits no nontrivial finite étale covers in characteristic 0. We present a concise reduction to $\mathbb{C}$ and invoke the comparison between finite étale covers and finite topological coverings (Riemann existence) [4, Exposé XII], [13, III.4].

Stage 7: Final birational and quasi-finite reduction

Triviality of the étale cover implies σ is an isomorphism, hence f is birational. Since $\det DF \in k^\times$, the original map F is étale and thus quasi-finite. Zariski's Main Theorem then factors F as an open immersion followed by a finite morphism; the finite birational morphism

to the normal surface $\mathbb{A}^2$ is an isomorphism [6, Theorem 8.12.6], [17, Corollary (4.C)]. Therefore F is an automorphism.

The only genuinely non-formal input of the proof occurs in Stage 4, where valuation-theoretic degeneration forces a rank-one structure in the associated graded ring. All other steps rely solely on standard tools: Stein factorization, generic flatness, étale comparison, and Zariski's Main Theorem.

Chapter 2

The Normalized Graph

2.1 Graph Closure and Normalization

Projective Compactification

Let

$$F = (P, Q) : \mathbb{A}_k^2 \longrightarrow \mathbb{A}_k^2$$

be a polynomial morphism over an algebraically closed field k of characteristic 0.

We embed $\mathbb{A}^2$ into $\mathbb{P}^2$ via the standard open chart $z \neq 0$. Let

$$\Gamma_F^{\text{aff}} \subset \mathbb{A}^2 \times \mathbb{A}^2$$

be the affine graph of F . We define the projective graph closure

$$\Gamma_F \subset \mathbb{P}^2 \times \mathbb{P}^2$$

to be the Zariski closure of Γ_F^{aff} .

Since $\mathbb{P}^2 \times \mathbb{P}^2$ is projective, Γ_F is a projective surface over k ([8, II.4]).

Basic Properties of the Graph

Lemma 2.1. *The affine graph Γ_F^{aff} is irreducible. Hence its projective closure Γ_F is irreducible.*

Proof. The affine graph is isomorphic to $\mathbb{A}^2$ via projection to the first factor. Hence it is irreducible. Its Zariski closure inside $\mathbb{P}^2 \times \mathbb{P}^2$ remains irreducible ([14, Chapter I, §1]). ■

Normalization

Let

$$\nu : X \longrightarrow \Gamma_F$$

be the normalization of Γ_F .

Since Γ_F is an irreducible projective surface, its normalization exists and is finite ([8, II.6.3], [15, Tag 035H]).

Thus X is a normal projective surface, and the morphism ν is finite and birational.

Projection to the Target

Let

$$\pi_2 : \Gamma_F \longrightarrow \mathbb{P}^2$$

be the second projection. Composing with normalization gives a morphism

$$\pi_2 : X \longrightarrow \mathbb{P}^2.$$

Since Γ_F is closed in the projective variety $\mathbb{P}^2 \times \mathbb{P}^2$, the projection π_2 is proper ([8, II.4.8]).

The Affine Part

Let

$$X_0 := \pi_2^{-1}(\mathbb{A}^2) \subset X.$$

Define

$$f := \pi_2|_{X_0} : X_0 \longrightarrow \mathbb{A}^2.$$

Lemma 2.2. *The restriction of f to the open subset corresponding to the original affine graph coincides with the original morphism F .*

Proof. Over the open subset where both projective coordinates correspond to affine charts, Γ_F coincides with the affine graph. Normalization is an isomorphism over the smooth locus ([8, II.8]), and the projection recovers F . ■

Dominance

Lemma 2.3. *The morphism*

$$f : X_0 \rightarrow \mathbb{A}^2$$

is dominant.

Proof. Since $\det DF \in k^\times$, the morphism F is étale and therefore smooth of relative dimension 0 ([8, III.10]). In particular, F is dominant. As f extends F on a dense open subset, f is also dominant. ■

2.2 The Affine Part

Restriction to the Affine Target Chart

Let

$$\pi_2 : X \longrightarrow \mathbb{P}^2$$

be the morphism induced by the second projection $\mathbb{P}^2 \times \mathbb{P}^2 \rightarrow \mathbb{P}^2$. Since X is finite over Γ_F and Γ_F is closed in $\mathbb{P}^2 \times \mathbb{P}^2$, the morphism π_2 is proper ([8, II.4.8]).

Let $\mathbb{A}^2 \subset \mathbb{P}^2$ be the standard affine chart. We define

$$X_0 := \pi_2^{-1}(\mathbb{A}^2).$$

Since $\mathbb{A}^2$ is open in $\mathbb{P}^2$, the subset X_0 is open in X . In particular, X_0 is a normal quasi-projective surface.

We set

$$f := \pi_2|_{X_0} : X_0 \longrightarrow \mathbb{A}^2.$$

Compatibility with the Original Morphism

Lemma 2.4. *There exists a dense open subset $U \subset X_0$ such that*

$$f|_U = F$$

under the natural identification $U \cong \mathbb{A}^2$.

Proof. Over the open subset where both factors of $\mathbb{P}^2 \times \mathbb{P}^2$ are affine, the projective graph Γ_F coincides with the affine graph of F . Since normalization is an isomorphism over the smooth locus ([8, II.8.3]), and the affine graph is smooth, the normalization map induces an isomorphism over this dense open subset. On this open set, the projection π_2 coincides with F . ■

Dominance and Generic Separability

Lemma 2.5. *The morphism*

$$f : X_0 \rightarrow \mathbb{A}^2$$

is dominant and generically separable.

Proof. Since $\det DF \in k^\times$, the morphism F is étale by the Jacobian criterion ([8, III.10], [2, Theorem 17.11]). In particular, F is dominant and generically separable. Because f extends F over a dense open subset, f is also dominant and generically separable. ■

Lemma 2.6 (Normal Surface Structure). The scheme X_0 is an integral normal surface. In particular, every codimension one point of X_0 corresponds to a discrete valuation ring.

Proof. The scheme X is the normalization of the irreducible projective surface Γ_F , hence is integral and normal [8, II.6.3]. Since X_0 is an open subset of X , it remains integral and normal.

Because X_0 is a normal Noetherian integral scheme of dimension two, every codimension one local ring is a discrete valuation ring [8, II.6.11]. ■

Chapter 3

Stein Factorization

3.1 Existence of Stein Factorization

Properness and Finiteness Conditions

Recall that

$$f = \pi_2|_{X_0} : X_0 \longrightarrow \mathbb{A}^2$$

is the restriction of the proper morphism

$$\pi_2 : X \longrightarrow \mathbb{P}^2.$$

Since X is projective and π_2 is induced by a projection from a projective variety, π_2 is proper ([8, II.4.8]). Therefore f is proper as a restriction of a proper morphism to the inverse image of an open subset ([15, Tag 01W6]).

Moreover, X_0 is normal and of finite type over k , and $\mathbb{A}^2$ is affine and normal.

Stein Factorization

Lemma 3.1 (Properness of f). *The morphism*

$$f : X_0 \longrightarrow \mathbb{A}^2$$

is proper.

Proof. The morphism

$$\pi_2 : X \longrightarrow \mathbb{P}^2$$

is proper because X is projective over k and π_2 is induced by projection from $\mathbb{P}^2 \times \mathbb{P}^2$ ([8, II.4.8]).

Since $\mathbb{A}^2 \subset \mathbb{P}^2$ is open, we have

$$X_0 = \pi_2^{-1}(\mathbb{A}^2).$$

The restriction of a proper morphism to the inverse image of an open subset remains proper ([15, Tag 01W6]).

Therefore

$$f = \pi_2|_{X_0}$$

is proper. ■

Proposition 3.2 (Stein Factorization). *The morphism f admits a factorization*

$$f = \sigma \circ \rho,$$

where

$$\rho : X_0 \longrightarrow Y_0$$

is a proper morphism with connected fibers and

$$\sigma : Y_0 \longrightarrow \mathbb{A}^2$$

is finite.

Proof. Since f is proper and of finite type between Noetherian schemes, the Stein factorization applies ([8, III.11.5], [3, Corollary 4.3.3]).

Define

$$Y_0 := \operatorname{Spec} f_* \mathcal{O}_{X_0}.$$

Then the natural morphism

$$\rho : X_0 \longrightarrow Y_0$$

has connected fibers, and the induced morphism

$$\sigma : Y_0 \longrightarrow \mathbb{A}^2$$

is finite. ■

Irreducibility and Normality

Lemma 3.3. *The scheme Y_0 is irreducible and normal.*

Proof. Since X_0 is irreducible and ρ has connected fibers, Y_0 is irreducible. Furthermore, Y_0 is finite over the normal scheme $\mathbb{A}^2$. As a finite algebra over a normal domain that is integrally closed in its function field, Y_0 is normal ([15, Tag 035H], [5, IV.6.14]). ■

Function Field Identification

Lemma 3.4. *One has*

$$k(Y_0) = k(X_0).$$

In particular, ρ is birational.

Proof. By construction,

$$Y_0 = \operatorname{Spec} f_* \mathcal{O}_{X_0}.$$

Since f is dominant, the induced extension of function fields

$$k(\mathbb{A}^2) \subset k(X_0)$$

is finite. The scheme Y_0 is obtained by taking the integral closure of $\mathbb{A}^2$ in $k(X_0)$ ([5, IV.6.11]). Hence its function field equals $k(X_0)$. Therefore ρ induces an isomorphism of function fields and is birational. ■

3.2 Function Field Identification

Equality of Function Fields

Lemma 3.5. *One has*

$$k(Y_0) = k(X_0).$$

In particular, the morphism

$$\rho : X_0 \longrightarrow Y_0$$

is birational.

Proof. Since $f : X_0 \rightarrow \mathbb{A}^2$ is dominant, the induced map of function fields

$$k(\mathbb{A}^2) \hookrightarrow k(X_0)$$

is injective and finite.

By definition of Stein factorization,

$$Y_0 = \operatorname{Spec}(f_*\mathcal{O}_{X_0}),$$

and $\sigma : Y_0 \rightarrow \mathbb{A}^2$ is finite ([8, III.11.5], [3, Cor. 4.3.3]).

Over an affine open $V = \operatorname{Spec} A \subset \mathbb{A}^2$, the algebra

$$\Gamma(\sigma^{-1}(V), \mathcal{O}_{Y_0})$$

is equal to the integral closure of A in the total ring of fractions $k(X_0)$ ([5, IV.6.11], [15, Tag 035H]).

In particular, Y_0 is obtained by taking the normalization of $\mathbb{A}^2$ in the field $k(X_0)$. Hence the function field of Y_0 coincides with $k(X_0)$.

Therefore ρ induces an isomorphism on function fields and is birational. ■

Chapter 4

Degree Control

4.1 Generic Finite Flatness

Function Field Degree

Let

$$d = [k(Y_0) : k(\mathbb{A}^2)]$$

be the degree of the induced extension of function fields. Since $\sigma : Y_0 \rightarrow \mathbb{A}^2$ is finite and dominant, this degree equals the generic rank of $\sigma_* \mathcal{O}_{Y_0}$ ([5, IV.2.5], [15, Tag 00NX]).

Generic Freeness

Lemma 4.1. *There exists a dense open subset $S^\circ \subset \mathbb{A}^2$ such that*

$$\sigma^{-1}(S^\circ) \longrightarrow S^\circ$$

is finite flat of rank d .

Proof. Since σ is finite, the pushforward $\sigma_* \mathcal{O}_{Y_0}$ is a coherent $\mathcal{O}_{\mathbb{A}^2}$ -module ([8, III.4.2]).

By the generic freeness theorem ([15, Tag 00NX], [12, Theorem 24.1]), there exists a dense open subset $S^\circ \subset \mathbb{A}^2$ such that

$$\sigma_* \mathcal{O}_{Y_0}|_{S^\circ}$$

is locally free of constant rank.

The rank equals the dimension of the fiber over the generic point, which is precisely

$$d = [k(Y_0) : k(\mathbb{A}^2)]$$

([5, IV.2.5]).

Thus σ is finite flat of rank d over S° . ■

4.2 Finite Étale Restriction

Étaleness on the Affine Chart

Recall that on the dense open subset

$$U \subset X_0$$

identified with the original affine graph, the morphism f coincides with

$$F : \mathbb{A}^2 \rightarrow \mathbb{A}^2.$$

Since $\det DF \in k^\times$, the Jacobian criterion implies that F is smooth of relative dimension 0 ([8, III.10], [2, Theorem 17.11]). Hence F is étale ([7, IV.17.5.1], [15, Tag 00U3]).

Therefore

$$f|_U : U \rightarrow \mathbb{A}^2$$

is étale, and in particular quasi-finite.

Application of Zariski Main Theorem

Lemma 4.2. *There exists a dense open subset*

$$S^{\circ\circ} \subset \mathbb{A}^2$$

such that

$$f|_{U \cap f^{-1}(S^{\circ\circ})}$$

is finite étale of degree d .

Proof. Since $f|_U$ is quasi-finite and separated, Zariski Main Theorem implies that there exists a factorization

$$U \xhookrightarrow{j} \tilde{U} \rightarrow \mathbb{A}^2,$$

where j is an open immersion and $\tilde{U} \rightarrow \mathbb{A}^2$ is finite ([6, Theorem 8.12.6], [8, Theorem 4.4]).

Shrinking the target if necessary, there exists a dense open

$$S^{\circ\circ} \subset \mathbb{A}^2$$

such that the restriction

$$f|_{U \cap f^{-1}(S^{\circ\circ})}$$

is finite.

Since étaleness is stable under base change and restriction ([7, IV.17.7.3], [15, Tag 02GU]), this restriction remains étale.

It remains to identify the degree. Over the generic point η of $\mathbb{A}^2$, the fiber has cardinality equal to the degree of the function field extension

$$[k(Y_0) : k(\mathbb{A}^2)] = d.$$

Because U is dense in X_0 and $k(U) = k(X_0) = k(Y_0)$, the generic fiber degree of $f|_U$ equals d ([5, IV.2.5]).

Hence

$$f|_{U \cap f^{-1}(S^{\circ\circ})}$$

is finite étale of degree d . ■

Chapter 5

Elimination of Horizontal Boundary Curves

5.1 Statement of U-HIT

Geometric Setup

Recall that

$$f : X_0 \longrightarrow \mathbb{A}^2$$

is proper and dominant, with Stein factorization

$$f = \sigma \circ \rho,$$

where

$$\rho : X_0 \rightarrow Y_0 \quad \text{has connected fibers,} \quad \sigma : Y_0 \rightarrow \mathbb{A}^2 \quad \text{is finite}$$

([3, III.11.5]).

Let

$$U \subset X_0$$

be the dense open subset identified with the affine graph of the original map, so that $f|_U = F$.

Let

$$d = [k(Y_0) : k(\mathbb{A}^2)]$$

be the function field degree.

From Chapter 4 we have dense open subsets

$$S^\circ, S^{\circ\circ} \subset \mathbb{A}^2$$

such that:

- σ is finite flat of rank d over S° ([5, IV.2.5]);
- $f|_U$ is finite étale of degree d over $S^{\circ\circ}$ ([6, Theorem 8.12.6]; étaleness stable under restriction [7, IV.17.7.3]).

Fiber Length Identity

Lemma 5.1 (Fiber length over S°). *For every closed point $s \in S^\circ$ one has*

$$\text{length}(f^{-1}(s)) = d.$$

Proof. Since σ is finite flat of constant rank d over S° , the $\mathcal{O}_{S^\circ}$ -module

$$\sigma_* \mathcal{O}_{Y_0}$$

is locally free of rank d ([5, IV.2.5]).

By Stein factorization,

$$f_* \mathcal{O}_{X_0} = \sigma_* \mathcal{O}_{Y_0}.$$

Hence over S° , the sheaf $f_* \mathcal{O}_{X_0}$ is locally free of rank d .

For a closed point $s \in S^\circ$ with residue field $\kappa(s)$, base change yields

$$\mathcal{O}_{f^{-1}(s)} \cong (f_* \mathcal{O}_{X_0}) \otimes_{\mathcal{O}_{\mathbb{A}^2}} \kappa(s).$$

Since the latter is a $\kappa(s)$ -vector space of dimension d , the scheme-theoretic fiber has length d . ■

Lemma 5.2 (Contribution from the affine chart). *For every closed point $s \in S^{\circ\circ}$, the fiber*

$$f^{-1}(s) \cap U$$

consists of exactly d reduced points. In particular,

$$\text{length}(f^{-1}(s) \cap U) = d.$$

Proof. Over $S^{\circ\circ}$, the morphism

$$f|_U : U \rightarrow \mathbb{A}^2$$

is finite étale of degree d ([6, Theorem 8.12.6]).

For a closed point $s \in S^{\circ\circ}$, finite étale morphisms have reduced fibers of cardinality equal to the degree ([7, IV.17.5.1]). Hence the fiber consists of d distinct reduced points. ■

U-HIT

Theorem 5.3 (U-HIT). *There exists no irreducible curve*

$$D \subset X_0 \setminus U$$

such that $\dim f(D) = 1$.

Proof. Assume for contradiction that such a curve D exists. Let

$$C := \overline{f(D)} \subset \mathbb{A}^2.$$

Since f is proper, C is closed ([8, II.4.8]).

Because $D \rightarrow C$ is dominant, for a general closed point $s \in C$ we have

$$D \cap f^{-1}(s) \neq \emptyset.$$

Choose

$$s \in C \cap S^\circ \cap S^{\circ\circ}.$$

This intersection is nonempty since all three sets are dense open.

For this s we have:

$$\text{length}(f^{-1}(s)) = d \quad (\text{Lemma 5.1})$$

and

$$\text{length}(f^{-1}(s) \cap U) = d \quad (\text{Lemma 5.2}).$$

Since U is open in X_0 , the scheme-theoretic fiber decomposes as

$$f^{-1}(s) = (f^{-1}(s) \cap U) \sqcup (f^{-1}(s) \cap (X_0 \setminus U)).$$

Length is additive on finite disjoint unions of zero-dimensional schemes. Therefore

$$\text{length}(f^{-1}(s)) = \text{length}(f^{-1}(s) \cap U) + \text{length}(f^{-1}(s) \cap (X_0 \setminus U)).$$

Substituting the identities above yields

$$d = d + \text{length}(f^{-1}(s) \cap (X_0 \setminus U)),$$

hence

$$\text{length}(f^{-1}(s) \cap (X_0 \setminus U)) = 0.$$

Thus

$$f^{-1}(s) \cap (X_0 \setminus U) = \emptyset,$$

contradicting

$$D \cap f^{-1}(s) \neq \emptyset.$$

Therefore no such curve D exists. ■

Remark 5.4 (Pure Fiber-Length Argument). The proof of Theorem 5.3 does not use any global quasi-finiteness assumption on f or on σ . In particular, no prior control on the behavior of fibers outside the dense open subsets is required.

The argument relies solely on:

- the flat fiber-length identity for σ over S° ,
- the finite étale fiber structure of $f|_U$ over $S^{\circ\circ}$,
- and additivity of length for zero-dimensional schemes.

Thus U-HIT is a purely fiber-length theoretic separation argument, independent of any global quasi-finiteness or birationality input. In particular, no structural property of the Stein factor Y_0 beyond finite flatness over S° enters the argument.

Chapter 6

Valuation-Theoretic Analysis

6.1 Local DVR Setup

Prime Divisors and Discrete Valuations

Let

$$D \subset X_0$$

be an irreducible curve. Since X_0 is a normal surface, every codimension one point corresponds to a discrete valuation ring ([8, II.6.11], [15, Tag 0A7G]).

Let

$$\eta_D$$

denote the generic point of D . Then

$$R := \mathcal{O}_{X_0, \eta_D}$$

is a discrete valuation ring (DVR).

Lemma 6.1 (Nonnegativity of valuations for regular functions). *Let $D \subset X_0$ be a prime divisor and let*

$$v = \text{ord}_D$$

be the associated valuation on $k(X_0)$. Since $P, Q \in \Gamma(X_0, \mathcal{O}_{X_0})$ are regular functions, one has

$$v(P) \geq 0 \quad \text{and} \quad v(Q) \geq 0.$$

Proof. A regular function on the normal variety X_0 has no pole along any prime divisor. Equivalently, its order along every prime divisor is nonnegative. ■

We denote its valuation by

$$v = \text{ord}_D.$$

Let t be a uniformizer of R .

Valuation of the Coordinate Functions

Since $f : X_0 \rightarrow \mathbb{A}^2$ extends the polynomial map

$$F = (P, Q),$$

the functions P and Q may be viewed as elements of the function field $k(X_0)$.

Via the natural inclusion

$$R \hookrightarrow \text{Frac}(R) = k(X_0),$$

we regard $P, Q \in R$ as elements of the fraction field whenever convenient.

Define

$$a := v(P), \quad b := v(Q).$$

Then we may write in R :

$$P = t^a u, \quad Q = t^b v,$$

where

$$u, v \in R^\times.$$

Residue Field and Units

Let

$$\kappa(D) = R/(t)$$

be the residue field.

Lemma 6.2 (Constant Field of the Residue Function Field). *Let $D \subset X_0$ be an irreducible curve. Then the algebraic closure of k inside $\kappa(D)$ equals k . In particular, the constant field of $\kappa(D)$ is k .*

Proof. Since X_0 is of finite type over the algebraically closed field k , the curve D is integral and of finite type over k [14, Chapter I, §4].

Hence $\kappa(D)$ is a finitely generated field extension of k of transcendence degree one:

$$\text{trdeg}_k \kappa(D) = 1.$$

Let $\alpha \in \kappa(D)$ be algebraic over k . Since k is algebraically closed, every element algebraic over k already lies in k . Thus $\alpha \in k$.

Therefore the algebraic closure of k in $\kappa(D)$ equals k . ■

Since X_0 is of finite type over the algebraically closed field k , the field $\kappa(D)$ is a finitely generated extension of k of transcendence degree one ([14, Chapter I, §4]).

Denote by

$$u_0, v_0 \in \kappa(D)^\times$$

the residue classes of u and v .

These residue units will encode the leading terms of P and Q in the associated graded ring of the valuation filtration.

6.2 Associated Graded Ring

Valuation Filtration

Let

$$v = \text{ord}_D$$

be the discrete valuation associated with the prime divisor D . The valuation induces a decreasing filtration on the function field $k(X_0)$ given by

$$\mathcal{F}^m := \{x \in k(X_0) \mid v(x) \geq m\}.$$

Restricting to the valuation ring

$$R = \mathcal{O}_{X_0, \eta_D},$$

we obtain a filtration by powers of the maximal ideal:

$$\mathcal{F}^m R = (t^m),$$

where t is a uniformizer.

Associated Graded Ring

The associated graded ring of R with respect to the valuation filtration is

$$\mathrm{gr}_v(R) = \bigoplus_{m \geq 0} \mathcal{F}^m R / \mathcal{F}^{m+1} R.$$

Since R is a DVR with residue field $\kappa(D)$, there is a natural identification

$$\mathrm{gr}_v(R) \cong \kappa(D)[T],$$

where T corresponds to the class of t ([17, Chapter VI, §1]).

Via this identification

$$\mathrm{gr}_v(R) \cong \kappa(D)[T],$$

we regard F_0 and G_0 as elements of the Laurent ring

$$\kappa(D)[T, T^{-1}]$$

without further distinction.

Thus the associated graded ring is a polynomial ring over the residue field.

Initial Forms

Let

$$a = v(P), \quad b = v(Q).$$

Write

$$P = t^a u, \quad Q = t^b v,$$

with $u, v \in R^\times$. Let

$$u_0, v_0 \in \kappa(D)^\times$$

be their residue classes.

The initial forms of P and Q in the graded ring are then

$$F_0 := u_0 T^a, \quad G_0 := v_0 T^b$$

as elements of $\mathrm{gr}_v(R)$.

These initial forms encode the leading terms of P and Q with respect to the valuation v . They will serve as the input for the degeneration argument in the next chapter.

Chapter 7

Rees Degeneration

7.1 Construction

The Valuation and its Rees Algebra

Let

$$v = \text{ord}_D$$

be the discrete valuation associated to the prime divisor

$$D \subset X_0.$$

Let

$$R = \mathcal{O}_{X_0, \eta_D}$$

be the corresponding discrete valuation ring with uniformizer t and residue field $\kappa = \kappa(D)$.

The valuation defines the filtration

$$\mathcal{F}^m R := \{x \in R \mid v(x) \geq m\} = (t^m) \subset R \quad (m \geq 0).$$

The associated Rees algebra is

$$\mathcal{R} := \bigoplus_{m \geq 0} \mathcal{F}^m R \cdot s^m \subset R[s],$$

where s is an indeterminate.

Since $\mathcal{F}^m R = (t^m)$, we have

$$\mathcal{R} = \bigoplus_{m \geq 0} (t^m) s^m = R[ts] \subset R[s].$$

Thus $\mathcal{R}$ is the R -subalgebra of $R[s]$ generated by the single element ts .

Flatness

Lemma 7.1 (Flatness of the Rees algebra for a DVR filtration). *The Rees algebra $\mathcal{R}$ is flat over $k[s]$. Consequently the structure morphism*

$$\pi : \text{Spec}(\mathcal{R}) \longrightarrow \mathbb{A}_k^1$$

is flat.

Proof. Since R is a domain, the subring $\mathcal{R} = R[ts] \subset R[s]$ is torsion-free as a $k[s]$ -module. Because $k[s]$ is a principal ideal domain, torsion-free modules over $k[s]$ are flat ([12, Thm. 7.7]). ■

Fibers of the Rees Family

Let

$$\mathcal{X} := \operatorname{Spec}(\mathcal{R}).$$

The morphism

$$\pi : \mathcal{X} \rightarrow \mathbb{A}_k^1$$

is induced by the inclusion $k[s] \subset \mathcal{R}$.

Generic fiber. Localizing at $s \neq 0$ yields

$$\mathcal{R}[s^{-1}] = R[ts][s^{-1}] = R[s, s^{-1}],$$

since $t = (ts)/s$ after inverting s . Hence the generic fiber is

$$\mathcal{X}_\eta \simeq \operatorname{Spec}(R).$$

Special fiber. Reducing modulo s gives

$$\mathcal{R}/(s) = \bigoplus_{m \geq 0} (t^m)/(t^{m+1}),$$

which is the associated graded ring

$$\operatorname{gr}_v(R).$$

Thus the special fiber is

$$\mathcal{X}_0 = \operatorname{Spec}(\operatorname{gr}_v(R)).$$

Degeneration of the Coordinate Functions

Let $P, Q \in R$. Write

$$P = t^a u, \quad Q = t^b v$$

with $u, v \in R^\times$.

In the Rees algebra, define

$$\tilde{P} := s^a P = (ts)^a u, \quad \tilde{Q} := s^b Q = (ts)^b v.$$

These lie in $\mathcal{R}$ and are homogeneous of degree a and b with respect to the s -grading.

In the generic fiber $s \neq 0$, dividing by appropriate powers of s recovers the original elements P, Q in R .

In the special fiber $s = 0$, the images of $\tilde{P}, \tilde{Q}$ are precisely the initial forms

$$F_0 = u_0 T^a, \quad G_0 = v_0 T^b$$

in the associated graded ring

$$\operatorname{gr}_v(R) \cong \kappa[T, T^{-1}],$$

where T denotes the class of t modulo t^2 .

Thus the Rees family provides a flat degeneration of (P, Q) to their initial forms (F_0, G_0) .

7.2 Dimension Transport

Dicritical Divisors

Let

$$D \subset X_0$$

be an irreducible curve with generic point η_D . We call D *dicritical* if

$$\dim f(D) = 1.$$

Equivalently, the induced extension of function fields

$$k(\mathbb{A}^2) \hookrightarrow k(D)$$

has transcendence degree one over k ([14, Chapter I, §6]).

Let

$$R = \mathcal{O}_{X_0, \eta_D}$$

be the associated DVR with valuation $v = \text{ord}_D$.

Rees Degeneration

Let $\mathcal{R}$ be the Rees algebra of the valuation filtration constructed in Section 7.1. By Lemma 7.1,

$$\mathcal{R} \text{ is flat over } k[s].$$

Denote

$$\mathcal{X} = \text{Spec}(\mathcal{R}), \quad \pi : \mathcal{X} \rightarrow \mathbb{A}^1.$$

The fibers satisfy:

$$\mathcal{X}_\eta \simeq \text{Spec}(R), \quad \mathcal{X}_0 = \text{Spec}(\text{gr}_v(R)).$$

The morphism $f = (P, Q)$ induces a morphism

$$g : \mathcal{X} \rightarrow \mathbb{A}^2$$

via the Rees lifts $\tilde{P}, \tilde{Q}$.

The Image Algebra

Define the $k[s]$ -algebra

$$B := \text{im} \left(k[s][U, V] \longrightarrow \mathcal{R}, \quad U \mapsto \tilde{P}, \quad V \mapsto \tilde{Q} \right) \subset \mathcal{R}.$$

Finite type. Since $k[s][U, V]$ is finitely generated over $k[s]$ and B is its image,

$$B \cong k[s][U, V] / \ker(\varphi),$$

hence B is a finitely generated $k[s]$ -algebra.

Flatness.

Lemma 7.2 (Flatness of the image algebra). *The $k[s]$ -algebra*

$$B := \text{im}\left(k[s][U, V] \longrightarrow \mathcal{R}, U \mapsto \tilde{P}, V \mapsto \tilde{Q}\right) \subset \mathcal{R}$$

is flat over $k[s]$.

Proof. By Lemma 7.1 the Rees algebra $\mathcal{R}$ is flat over $k[s]$, hence torsion-free as a $k[s]$ -module.

Since $B \subset \mathcal{R}$ is a $k[s]$ -submodule of a torsion-free module, B is itself torsion-free.

Finally, $k[s]$ is a principal ideal domain (in particular a Prüfer domain), and over a Prüfer domain every torsion-free module is flat. Therefore B is flat over $k[s]$ (cf. [2, Thm. 7.7]). ■

Remark 7.3. Note that we do *not* claim that B is finitely generated as a $k[s]$ -module; this is unnecessary since torsion-free modules over the PID $k[s]$ are flat.

Dominance and Domain Structure of B

Lemma 7.4 (Lemma 7.2a). *The $k[s]$ -algebra B is an integral domain. Moreover, the structure morphism*

$$\text{Spec}(B) \longrightarrow \text{Spec}(k[s])$$

is dominant.

Proof. We first show that B is an integral domain.

By assumption, R is a domain. Indeed, R embeds into the discrete valuation ring $R[s]$, hence has no zero divisors. Since $B \subset R$ is a subring of a domain, it follows immediately that B itself is a domain.

We now prove dominance of the structure morphism. Let $\eta = (0) \in \text{Spec}(k[s])$ denote the generic point. Localizing at η corresponds to inverting s , hence the generic fiber is

$$B_\eta = B \otimes_{k[s]} k(s) \subset R[s, s^{-1}].$$

By construction, B_η contains the images of P and Q inside $\text{Frac}(R)$. Since D is dicritical, the field $k(P, Q)$ has transcendence degree at least 1 over k . In particular, the images of P and Q cannot both lie in k , and therefore $B_\eta \neq 0$.

Thus the generic fiber of $\text{Spec}(B) \rightarrow \text{Spec}(k[s])$ is nonempty. Consequently, the morphism is dominant. ■

Identification of the Fibers

Lemma 7.5. *The fibers of B satisfy:*

$$B_\eta \cong \text{im}(k(P, Q) \rightarrow \text{Frac}(R)),$$

$$B_0 \cong \kappa(D)[F_0, G_0] \subset \kappa(D)[T, T^{-1}].$$

Proof. Recall that

$$B = \text{im}(k[s][U, V] \longrightarrow \mathcal{R}), \quad U \mapsto \tilde{P}, \quad V \mapsto \tilde{Q}.$$

Generic fiber.

After inverting s , the Rees algebra satisfies

$$\mathcal{R}[s^{-1}] = R[s, s^{-1}].$$

Hence

$$B_\eta = B[s^{-1}] = \text{im}(k[s, s^{-1}][U, V] \rightarrow R[s, s^{-1}]).$$

Dividing $\tilde{P}, \tilde{Q}$ by the appropriate powers of s recovers the original elements P, Q inside $\text{Frac}(R)$. Therefore

$$B_\eta \cong \text{im}(k(P, Q) \rightarrow \text{Frac}(R)).$$

Special fiber.

Since B is defined as an image algebra, taking the quotient modulo (s) yields

$$B/(s) \cong \text{im}(k[U, V] \rightarrow \mathcal{R}/(s)).$$

But

$$\mathcal{R}/(s) = \text{gr}_v(R),$$

and the images of $\tilde{P}, \tilde{Q}$ reduce precisely to the initial forms

$$F_0, G_0.$$

Hence

$$B_0 = B/(s) = \text{im}(k[U, V] \rightarrow \text{gr}_v(R)) = \kappa(D)[F_0, G_0].$$

Since $\text{gr}_v(R) \cong \kappa(D)[T]$, this identifies B_0 as a subalgebra of

$$\kappa(D)[T, T^{-1}].$$

■

Dimension Transport

We now verify carefully that the dimension formula applies.

All dimensions in this chapter refer to Krull dimension.

Lemma 7.6. *Let B be a finitely generated flat $k[s]$ -algebra. Then*

$$\dim B_0 \geq \dim B_\eta.$$

Proof. The ring $k[s]$ is Noetherian, integral, and has Krull dimension 1.

Since B is finitely generated and flat over $k[s]$, the morphism

$$\text{Spec}(B) \rightarrow \text{Spec}(k[s])$$

is flat and of finite type.

Since B is a domain and $B_\eta \neq 0$ (Lemma 7.3), the structure morphism

$$\text{Spec}(B) \longrightarrow \text{Spec}(k[s])$$

is dominant.

The dimension formula for flat morphisms ([5, IV, Cor. 5.6.5]) yields

$$\dim B = \dim B_\eta + 1.$$

For any fiber B_t of a flat morphism,

$$\dim B_t \geq \dim B - \dim k[s] = \dim B - 1.$$

Hence

$$\dim B_0 \geq \dim B - 1 = \dim B_\eta.$$

■

Dicritical Case

Lemma 7.7. *If D is dicritical, then*

$$\dim B_\eta \geq 1.$$

Proof. Dicritical means that

$$\dim f(D) = 1.$$

At the generic point η_D , the rational functions P, Q cannot both be constant on D . Hence the field

$$k(P, Q) \subset \text{Frac}(R)$$

has transcendence degree at least one over k .

Since B_η is a finitely generated domain whose fraction field equals $k(P, Q)$, its Krull dimension equals

$$\text{trdeg}_k k(P, Q) \geq 1.$$

On the other hand, $k(P, Q) \subset k(X_0)$ and $\text{trdeg}_k k(X_0) = 2$, so necessarily $\text{trdeg}_k k(P, Q) = 1$. ■

Corollary 7.8. *If D is dicritical, then*

$$\dim \kappa(D)[F_0, G_0] \geq 1.$$

Proof. By Lemma 7.7 and Lemma 7.6,

$$\dim B_0 \geq 1.$$

Since

$$B_0 = \kappa(D)[F_0, G_0]$$

is a finitely generated algebra over a field, its Krull dimension equals the dimension of its largest irreducible component. ■

Chapter 8

Rank Dichotomy and Character Relation

8.1 Rank Dichotomy and Character Relation

From this point onward, all arguments take place entirely inside the associated graded ring of the DVR $R = \mathcal{O}_{X_0, \eta_D}$. No global geometry will re-enter until Chapter 10.

Laurent structure

Via the canonical identification

$$\Gamma_v(R) \cong \kappa(D)[T] \quad \text{and hence} \quad \text{Frac}(\Gamma_v(R)) \cong \kappa(D)(T),$$

we work in the Laurent domain

$$A := \kappa(D)[T, T^{-1}].$$

Fix

$$F_0 = u_0 T^a, \quad G_0 = v_0 T^b,$$

with $u_0, v_0 \in \kappa(D)^\times$ and $a, b \in \mathbb{Z}_{>0}$, and set

$$B := \kappa(D)[F_0, G_0] \subset \kappa(D)[T] \subset A.$$

Dimension hypothesis

Assume throughout this chapter that

$$\dim B = 1.$$

Consider the $\kappa(D)$ -algebra homomorphism

$$\varphi : \kappa(D)[U, V] \longrightarrow A, \quad U \mapsto F_0, \quad V \mapsto G_0,$$

and let $I := \ker(\varphi)$. Then $B \cong \kappa(D)[U, V]/I$.

Lemma 8.1 (Height-one kernel). *If $\dim B = 1$, then $\text{ht}(I) = 1$.*

Proof. Since $\kappa(D)[U, V]$ is a polynomial ring in two variables over a field, it has Krull dimension 2. Moreover,

$$\dim(\kappa(D)[U, V]/I) = \dim B = 1.$$

By Krull's dimension formula,

$$2 = \dim \kappa(D)[U, V] = \dim(\kappa(D)[U, V]/I) + \text{ht}(I) = 1 + \text{ht}(I).$$

Hence $\text{ht}(I) = 1$. ■

Lemma 8.2 (Laurent monomial independence). *The monomials $\{T^n\}_{n \in \mathbb{Z}}$ are $\kappa(D)$ -linearly independent in A .*

Proof. As a $\kappa(D)$ -vector space, $A = \kappa(D)[T, T^{-1}]$ has basis $\{T^n\}_{n \in \mathbb{Z}}$. ■

Theorem 8.3 (Rank dichotomy). *Assume $\dim B = 1$. Then there exist integers $(m, n) \neq (0, 0)$ and an element $c \in \kappa(D)^\times$ such that*

$$F_0^m G_0^n = c.$$

Equivalently, there exist $(m, n) \neq (0, 0)$ with

$$ma + nb = 0 \quad \text{and} \quad u_0^m v_0^n = c.$$

Proof. By Lemma 8.1, I has height one. Since $\kappa(D)[U, V]$ is a UFD, there exists a nonzero irreducible polynomial

$$H(U, V) = \sum_{(i,j)} \lambda_{ij} U^i V^j \in \kappa(D)[U, V]$$

such that $I = (H)$. Substituting $U = F_0$, $V = G_0$ gives in A :

$$0 = H(F_0, G_0) = \sum_{(i,j)} \lambda_{ij} u_0^i v_0^j T^{ai+bj}.$$

By Lemma 8.2, distinct exponents in T cannot cancel. Hence there exist two distinct pairs $(i_1, j_1) \neq (i_2, j_2)$ occurring with $\lambda_{i_\nu j_\nu} \neq 0$ such that

$$ai_1 + bj_1 = ai_2 + bj_2.$$

Set $m := i_1 - i_2$ and $n := j_1 - j_2$. Then $(m, n) \neq (0, 0)$ and $ma + nb = 0$.

Restricting $H(F_0, G_0) = 0$ to the common T -degree $ai_1 + bj_1$ yields

$$\lambda_{i_1 j_1} u_0^{i_1} v_0^{j_1} + \lambda_{i_2 j_2} u_0^{i_2} v_0^{j_2} = 0,$$

hence (in the field $\kappa(D)$)

$$u_0^m v_0^n = -\frac{\lambda_{i_2 j_2}}{\lambda_{i_1 j_1}} \in \kappa(D)^\times.$$

Finally,

$$F_0^m G_0^n = (u_0 T^a)^m (v_0 T^b)^n = u_0^m v_0^n T^{ma+nb} = u_0^m v_0^n =: c \in \kappa(D)^\times. \quad \text{■}$$

Remark 8.4 (Sign issue). The integers m, n obtained above need not be nonnegative. Thus the relation $F_0^m G_0^n = c$ is naturally an identity in the Laurent field $\kappa(D)(T)$ (and in the Laurent ring A), not necessarily inside $\kappa(D)[T]$.

Remark 8.5 (What this does *not* give). The conclusion $u_0^m v_0^n = c \in \kappa(D)^\times$ alone does *not* force $u_0, v_0 \in k^\times$ (even when $c \in k^\times$). Any descent to $k^\times$ requires additional input and is handled separately in §8.2.

8.2 Character Relation

From Rank Dichotomy to a multiplicative relation

Retain the hypotheses and notation of Theorem 8.3. Thus there exist integers $(m, n) \neq (0, 0)$ and an element $c \in \kappa(D)^\times$ such that

$$F_0^m G_0^n = c.$$

Writing

$$F_0 = u_0 T^a, \quad G_0 = v_0 T^b \quad (u_0, v_0 \in \kappa(D)^\times, \ a, b \in \mathbb{Z}_{>0}),$$

we expand in $\kappa(D)(T)$:

$$F_0^m G_0^n = u_0^m v_0^n T^{ma+nb}.$$

Since the left-hand side equals $c \in \kappa(D)^\times$ (i.e. has no T -dependence), we obtain the *weight relation*

$$ma + nb = 0$$

and hence the *residue relation*

$$u_0^m v_0^n = c \in \kappa(D)^\times.$$

Corollary 8.6 (Character relation over $\kappa(D)$). *There exist integers $(m, n) \neq (0, 0)$ and $c \in \kappa(D)^\times$ such that*

$$u_0^m v_0^n = c, \quad ma + nb = 0.$$

Constant field and what can (and cannot) be descended

Lemma 8.7 (Constant field of $\kappa(D)$). *Let D be an irreducible curve on X_0 . Then the algebraic closure of k inside $\kappa(D)$ equals k .*

Proof. Since X_0 is of finite type over the algebraically closed field k , the curve D is integral and of finite type over k [14, Ch. I, §4]. Hence $\text{trdeg}_k \kappa(D) = 1$. If $\alpha \in \kappa(D)$ is algebraic over k , then its minimal polynomial over k splits in $k[T]$ because k is algebraically closed, so $\alpha \in k$. ■

Remark 8.8 (No false descent). From $u_0^m v_0^n \in k^\times$ one must *not* conclude $u_0, v_0 \in k^\times$: in a function field $\kappa(D)$ it is possible that two nonconstant units satisfy a multiplicative relation with constant product (e.g. in $k(t)$ one has $t \cdot t^{-1} = 1$). Thus Corollary 8.6 alone does not yield strict positivity via translation, and we do not attempt such a reduction here.

A sufficient criterion for descent of the constant

The only descent we will ever need later is the descent of the *constant* $c = u_0^m v_0^n$ to $k^\times$ (not the constancy of u_0 or v_0 individually). This is guaranteed once the multiplicative relation comes from a polynomial relation with coefficients in k .

Lemma 8.9 (Descent of c from a k -relation). *Assume there exists a nonzero polynomial $H(U, V) \in k[U, V]$ such that*

$$H(F_0, G_0) = 0 \quad \text{in } \kappa(D)[T, T^{-1}],$$

and that $\dim \kappa(D)[F_0, G_0] = 1$. Then one may choose $(m, n) \neq (0, 0)$ in Theorem 8.3 such that

$$u_0^m v_0^n = c \in k^\times \quad \text{and} \quad ma + nb = 0.$$

Proof. Write $H(U, V) = \sum_{(i,j)} \lambda_{ij} U^i V^j$ with $\lambda_{ij} \in k$ and substitute $U = F_0 = u_0 T^a$, $V = G_0 = v_0 T^b$:

$$0 = H(F_0, G_0) = \sum_{(i,j)} \lambda_{ij} u_0^i v_0^j T^{ai+bj}.$$

By Laurent monomial independence (Lemma 8.2), there exist two distinct exponent pairs $(i_1, j_1) \neq (i_2, j_2)$ occurring with $\lambda_{i_\nu j_\nu} \neq 0$ such that

$$ai_1 + bj_1 = ai_2 + bj_2.$$

Set $m := i_1 - i_2$ and $n := j_1 - j_2$. Then $(m, n) \neq (0, 0)$ and $ma + nb = 0$. Looking at the coefficient of the common Laurent monomial $T^{ai_1+bj_1}$ gives

$$\lambda_{i_1 j_1} u_0^{i_1} v_0^{j_1} + \lambda_{i_2 j_2} u_0^{i_2} v_0^{j_2} = 0,$$

hence

$$u_0^m v_0^n = -\frac{\lambda_{i_2 j_2}}{\lambda_{i_1 j_1}} \in k^\times.$$

■

Corollary 8.10 (Character relation over k (when available)). *Under the hypotheses of Lemma 8.9, there exist integers $(m, n) \neq (0, 0)$ and a constant $c \in k^\times$ such that*

$$u_0^m v_0^n = c, \quad ma + nb = 0.$$

Chapter 9

Wedge Strictness

9.1 Valuation of the Jacobian Form

Fix a prime divisor

$$D \subset X_0$$

with valuation

$$v = \text{ord}_D, \quad R = \mathcal{O}_{X_0, \eta_D}$$

the corresponding DVR with uniformizer t and residue field

$$\kappa = \kappa(D) = R/(t).$$

Since X_0 is a normal surface, R is a DVR at every codimension-one point [8, II.6.11].

Valuation Setup and Character Relation

Write

$$P = t^a u, \quad Q = t^b v$$

with $a = v(P)$, $b = v(Q)$ and $u, v \in R^\times$. Let $u_0, v_0 \in \kappa^\times$ be the residue classes.

Assume the *character relation*: there exist integers $(m, n) \neq (0, 0)$ and $c \in \kappa^\times$ such that

$$u_0^m v_0^n = c, \quad ma + nb = 0.$$

We assume only

$$a + b > 0 \quad (\text{equivalently: } a > 0 \text{ or } b > 0),$$

which is the input needed in Chapter 10.

A reduction lemma in Ω^1

We use the standard exact sequence for the quotient $R \rightarrow \kappa$:

$$(t)/(t)^2 \longrightarrow \Omega_{R/k}^1 \otimes_R \kappa \longrightarrow \Omega_{\kappa/k}^1 \longrightarrow 0.$$

In particular, if $d(\bar{x}) = 0$ in $\Omega_{\kappa/k}^1$ for $\bar{x} \in \kappa$ the residue of $x \in R$, then the class of dx in $\Omega_{R/k}^1 \otimes \kappa$ lies in the image of $(t)/(t)^2$, hence is a κ -multiple of dt .

Case 1: $a > 0$ and $b > 0$

In $\Omega_{R/k}^1$ we have

$$dP = t^a du + at^{a-1}u dt, \quad dQ = t^b dv + bt^{b-1}v dt.$$

Thus

$$dP \wedge dQ = t^{a+b-1} (t du \wedge dv + bu du \wedge dt - av dv \wedge dt).$$

Modulo (t) , the term $t du \wedge dv$ vanishes, and the coefficient of t^{a+b-1} is governed by

$$(bu_0 du_0 - av_0 dv_0) \wedge dt.$$

Differentiating $u_0^m v_0^n = c$ in $\Omega_{\kappa/k}^1$ gives

$$m \frac{du_0}{u_0} + n \frac{dv_0}{v_0} = 0 \iff mu_0 du_0 + nv_0 dv_0 = 0.$$

Since $ma + nb = 0$ and $(m, n) \neq (0, 0)$, the kernel of the nonzero functional $(x, y) \mapsto mx + ny$ in $\mathbb{Q}^2$ is one-dimensional and contains both $(a, b) \neq (0, 0)$ and (m, n) . Hence there exists $\lambda \in \mathbb{Q}^\times$ with

$$(a, b) = \lambda(m, n).$$

Multiplying $mu_0 du_0 + nv_0 dv_0 = 0$ by λ yields

$$au_0 du_0 + bv_0 dv_0 = 0,$$

equivalently

$$bu_0 du_0 - av_0 dv_0 = 0.$$

Therefore the coefficient at valuation $a + b - 1$ vanishes, and we conclude

$$\text{ord}_D(dP \wedge dQ) \geq a + b.$$

Case 2: $a > 0$ and $b = 0$

Here $Q = v$ is a unit in R . The weight relation $ma + nb = 0$ becomes $ma = 0$, hence $m = 0$. Thus $n \neq 0$ and the character relation reduces to

$$v_0^n = c \in k^\times.$$

Differentiating gives

$$nv_0^{n-1} dv_0 = 0 \Rightarrow dv_0 = 0 \text{ in } \Omega_{\kappa/k}^1.$$

By the exact sequence above, this implies that dv maps to a κ -multiple of dt in $\Omega_{R/k}^1 \otimes \kappa$, hence

$$dt \wedge dv \equiv 0 \text{ in } \Omega_{R/k}^2 \otimes \kappa.$$

Now

$$dP = t^a du + at^{a-1}u dt, \quad dQ = dv,$$

so

$$dP \wedge dQ = t^a du \wedge dv + at^{a-1}u dt \wedge dv = t^{a-1} (t du \wedge dv + au dt \wedge dv).$$

Modulo (t) the term $t du \wedge dv$ vanishes, and the remaining coefficient is governed by $dt \wedge dv$, which vanishes modulo (t) by the paragraph above. Hence the valuation- $a - 1$ term cancels and we obtain

$$\text{ord}_D(dP \wedge dQ) \geq a.$$

Since $a > 0$, this implies $\text{ord}_D(dP \wedge dQ) > 0$.

Case 3: $a = 0$ and $b > 0$

This is symmetric to Case 2 and yields

$$\text{ord}_D(dP \wedge dQ) \geq b > 0.$$

Conclusion

Combining the three cases, we have proved:

Lemma 9.1 (Wedge Strictness). *Assume $a, b \geq 0$ and $a + b > 0$. If there exist integers $(m, n) \neq (0, 0)$ and $c \in k^\times$ such that*

$$u_0^m v_0^n = c, \quad ma + nb = 0,$$

then

$$\text{ord}_D(dP \wedge dQ) \geq a + b.$$

In particular, if $a + b > 0$ then $\text{ord}_D(dP \wedge dQ) > 0$.

Chapter 10

Elimination of Affine Branch Curves

10.1 Keller Identity and Dicritical Divisors

Let

$$f : X_0 \longrightarrow \mathbb{A}^2$$

be the morphism induced by $F = (P, Q)$, and let

$$f = \sigma \circ \rho$$

be its Stein factorization with

$$\rho : X_0 \rightarrow Y_0 \text{ birational,} \quad \sigma : Y_0 \rightarrow \mathbb{A}^2 \text{ finite.}$$

Keller Identity on the Affine Chart

Let

$$U \subset X_0$$

be the dense open subset identified with the affine graph of F . On U the morphism f coincides with

$$F = (P, Q) : \mathbb{A}^2 \rightarrow \mathbb{A}^2.$$

Hence on U we have the classical Keller identity

$$dP \wedge dQ = (\det DF) dx \wedge dy.$$

Since $\det DF \in k^\times$, the 2-form $dP \wedge dQ$ is nowhere vanishing on U . Equivalently, it is a unit in the invertible sheaf $\Omega_{U/k}^2$.

Because X_0 is normal, every prime divisor $D \subset X_0$ corresponds to a discrete valuation $v_D = \text{ord}_D$ on the function field $k(X_0)$. The rational 2-form $dP \wedge dQ$ extends to an element of $\Omega_{k(X_0)/k}^2$.

If a prime divisor D meets U , then the generic point of D lies in U . Since $dP \wedge dQ$ is a unit at that point, it has neither zero nor pole there. Therefore

$$\text{ord}_D(dP \wedge dQ) = 0.$$

10.2 Branch Curves Lift to Dicritical Divisors

Lemma 10.1 (Branch curves lift). *Assume σ is not étale. Let*

$$C \subset \mathbb{A}^2$$

be an irreducible component of the branch locus of σ . Then there exists a prime divisor

$$D \subset X_0$$

such that

$$f(D) = C.$$

In particular, D is dicritical.

Proof. Since σ is finite and $\mathbb{A}^2$ is regular of dimension 2, purity of the branch locus implies that every irreducible component of the branch locus has codimension one. Hence C is a curve.

Let $E \subset Y_0$ be an irreducible component of $\sigma^{-1}(C)$ that dominates C . Because Y_0 is a normal surface, the local ring $\mathcal{O}_{Y_0, \eta_E}$ is a discrete valuation ring [8, II.6.11].

The Stein factorization gives

$$k(X_0) = k(Y_0),$$

hence the valuation ord_E of $k(Y_0)$ induces a discrete valuation of $k(X_0)$. Since X_0 is normal, this valuation is realized by a unique prime divisor

$$D \subset X_0$$

[8, II.6.11].

By construction, the generic point of D maps to η_E . Therefore $\rho(D)$ is dense in E and

$$f(D) = \sigma(\rho(D)) = \sigma(E) = C.$$

Since $\dim C = 1$, we obtain $\dim f(D) = 1$, so D is dicritical. ■

10.3 Positive Valuations Along Branch Divisors

Lemma 10.2 (Branch divisors force $a + b > 0$). *Assume that $\sigma : Y_0 \rightarrow \mathbb{A}^2$ is not étale and let $C \subset \mathbb{A}^2$ be an irreducible component of the branch locus. Let $D \subset X_0$ be the prime divisor given by Lemma 10.1 with $f(D) = C$, and set $v := \text{ord}_D$. Then*

$$v(P) + v(Q) > 0.$$

Equivalently, at least one of $v(P), v(Q)$ is positive.

Proof. Let $R = \mathcal{O}_{X_0, \eta_D}$ be the DVR at the generic point of D .

Assume for contradiction that

$$v(P) = v(Q) = 0.$$

Then $P, Q \in R^\times$, hence their residues $P_0, Q_0 \in \kappa(D)^\times$ are nonzero.

Thus the generic point η_D maps into the principal open

$$\mathbb{A}_{PQ}^2 := \text{Spec } k[P, Q, (PQ)^{-1}] \subset \mathbb{A}^2,$$

i.e. into $\mathbb{G}_m^2$.

Over $\mathbb{A}_{PQ}^2$ the morphism f agrees on the dense open chart $U \simeq \mathbb{A}^2$ with the original Keller map $F = (P, Q)$. On U we have the identity

$$dP \wedge dQ = (\det DF) dx \wedge dy,$$

with $\det DF \in k^\times$. Hence $dP \wedge dQ$ is a nowhere vanishing 2-form on $U \cap f^{-1}(\mathbb{A}_{PQ}^2)$.

Since D dominates C and η_D lies over $\mathbb{A}_{PQ}^2$, the generic point of D lies in this locus. Thus $dP \wedge dQ$ is nonzero in

$$\Omega_{\text{Frac}(R)/k}^2,$$

and its class in

$$\Omega_{R/k}^2 \otimes_R \kappa(D)$$

is nonzero. Equivalently, the morphism f is unramified at η_D ([8, II.8.2]).

However, D maps onto a branch component C of the finite morphism $\sigma : Y_0 \rightarrow \mathbb{A}^2$. Hence σ is ramified at the generic point η_C of C . Since $f = \sigma \circ \rho$ and ρ is birational, ramification of σ along C implies that f is ramified at the generic point of D lying over η_C .

This contradicts the previous paragraph. Therefore $v(P)$ and $v(Q)$ cannot both be zero. ■

10.4 Valuation Contradiction

Assume for contradiction that

$$\sigma : Y_0 \rightarrow \mathbb{A}^2$$

is not étale.

Then its branch locus contains a curve. By Lemma 10.1 there exists a prime divisor

$$D \subset X_0$$

mapping onto a branch component C .

Set

$$a := v_D(P), \quad b := v_D(Q).$$

By Lemma 10.2,

$$a + b > 0.$$

By Chapter 9 (Wedge Strictness), the character relation implies

$$\text{ord}_D(dP \wedge dQ) \geq a + b.$$

Since $a + b > 0$, we obtain

$$\text{ord}_D(dP \wedge dQ) > 0.$$

On the other hand, D dominates a curve in $\mathbb{A}^2$, hence meets the affine chart U . On U the Keller identity holds:

$$dP \wedge dQ = (\det DF) dx \wedge dy,$$

with $\det DF \in k^\times$. Thus $dP \wedge dQ$ is a unit in codimension one along D , and

$$\text{ord}_D(dP \wedge dQ) = 0.$$

This is a contradiction.

Theorem 10.3. *The finite morphism*

$$\sigma : Y_0 \rightarrow \mathbb{A}^2$$

is étale.

Proof. We have excluded ramification in codimension one. Since σ is finite and Y_0 is normal, purity of the branch locus ([4, X.3.4]) implies that the ramification locus is empty. A finite unramified morphism is étale ([7, IV.17.4.2]). ■

Chapter 11

Triviality of Finite Étale Covers

Theorem 11.1. *Let k be an algebraically closed field of characteristic 0. Then*

$$\pi_1^t(\mathbb{A}_k^2) = 0.$$

Equivalently, every finite étale cover of $\mathbb{A}_k^2$ is trivial.

Proof. Step 1: Reduction to a finitely generated subfield.

Let $Y \rightarrow \mathbb{A}_k^2$ be a finite étale morphism. Since the morphism is of finite type, there exists a subfield $k_0 \subset k$ finitely generated over $\mathbb{Q}$ such that both $\mathbb{A}_k^2$ and Y descend to k_0 . That is, there exists a finite étale morphism

$$Y_0 \rightarrow \mathbb{A}_{k_0}^2$$

whose base change to k recovers Y .

Step 2: Embedding into $\mathbb{C}$.

Because k_0 is finitely generated over $\mathbb{Q}$, there exists an embedding

$$k_0 \hookrightarrow \mathbb{C}.$$

Base change along this embedding yields a finite étale morphism

$$Y_{\mathbb{C}} \rightarrow \mathbb{A}_{\mathbb{C}}^2.$$

Step 3: Comparison with complex topology.

The analytification of $\mathbb{A}_{\mathbb{C}}^2$ is

$$\mathbb{C}^2,$$

which is simply connected.

By the Riemann Existence Theorem (SGA 1, Exposé XII, Thm. 5.2), finite étale covers of $\mathbb{A}_{\mathbb{C}}^2$ correspond to finite topological covering spaces of $\mathbb{C}^2$.

Since $\mathbb{C}^2$ admits no nontrivial connected finite coverings, $Y_{\mathbb{C}}$ is a disjoint union of copies of $\mathbb{A}_{\mathbb{C}}^2$.

Step 4: Descent of triviality.

Because k is algebraically closed, extension of algebraically closed base fields does not change the étale fundamental group (SGA 1, Exposé XIII, Cor. 2.12).

Hence the triviality of the finite étale cover over $\mathbb{C}$ implies the triviality over k .

Therefore

$$\pi_1^t(\mathbb{A}_k^2) = 0.$$

■

Corollary 11.2. *The finite étale morphism*

$$\sigma : Y_0 \rightarrow \mathbb{A}^2$$

is an isomorphism.

Proof. Since σ is finite étale, the previous theorem implies that it is a disjoint union of copies of $\mathbb{A}^2$. Because Y_0 is irreducible, only one component can occur. Hence σ is an isomorphism. ■

Chapter 12

Final Step

12.1 Birationality

Reduction of the Stein Factorization

Recall the Stein factorization

$$f = \sigma \circ \rho$$

with

$$\rho : X_0 \rightarrow Y_0 \quad \text{and} \quad \sigma : Y_0 \rightarrow \mathbb{A}^2.$$

We have shown that σ is finite étale and, by triviality of finite étale covers of $\mathbb{A}^2$, an isomorphism.

Therefore

$$f = \rho$$

after identifying $Y_0 \cong \mathbb{A}^2$.

Birationality of f

Proposition 12.1. *The morphism*

$$f : X_0 \longrightarrow \mathbb{A}^2$$

is birational.

Proof. From the Stein construction we already know that

$$k(Y_0) = k(X_0).$$

Since σ is an isomorphism,

$$k(\mathbb{A}^2) = k(Y_0).$$

Hence

$$k(X_0) = k(\mathbb{A}^2).$$

Therefore f induces an isomorphism of function fields and is birational ([8, Chapter II, §6]). ■

12.2 Quasi-Finite Reduction

Étaleness of the Original Morphism

Recall that

$$F = (P, Q) : \mathbb{A}^2 \longrightarrow \mathbb{A}^2$$

satisfies

$$\det DF \in k^\times.$$

By the Jacobian criterion, F is smooth of relative dimension 0 ([8, III.10], [2, Theorem 17.11]). Hence F is étale ([7, IV.17.5.1], [15, Tag 00U3]).

Quasi-Finiteness

Since étale morphisms are locally quasi-finite ([15, Tag 02G4]), it follows that

$$F : \mathbb{A}^2 \longrightarrow \mathbb{A}^2$$

is quasi-finite.

Equivalently, every fiber of F consists of finitely many points, and the induced extension of function fields is finite.

Lemma 12.2 (Regular functions on a dense open of $\mathbb{A}_k^2$). *Let k be an algebraically closed field. Let $U \subset \mathbb{A}_k^2$ be a dense open subset. Then every regular function on U extends uniquely to a regular function on $\mathbb{A}_k^2$. In particular,*

$$\mathcal{O}(U) = k[x, y].$$

Proof. The affine plane $\mathbb{A}_k^2$ is normal, since it is the spectrum of the polynomial ring $k[x, y]$, which is a UFD and hence integrally closed ([8, II.6.11]).

Let $f \in \mathcal{O}(U)$. Since U is dense and $\mathbb{A}_k^2$ is normal, regular functions on U extend uniquely across codimension one subsets ([8, II.6.11]).

Therefore f extends to a global regular function on $\mathbb{A}_k^2$, which must be a polynomial in $k[x, y]$.

Hence $\mathcal{O}(U) = k[x, y]$. ■

Birational Étale Morphisms

From the previous section, we have shown that F is birational. Thus F is a birational quasi-finite morphism between integral schemes of finite type over k .

This reduction prepares the application of Zariski's Main Theorem in the final step.

12.3 Application of Zariski Main Theorem

Zariski Main Theorem

We have established that

$$F : \mathbb{A}^2 \longrightarrow \mathbb{A}^2$$

is birational and quasi-finite.

Zariski's Main Theorem in the quasi-finite form states that a separated quasi-finite morphism of finite type factors as

$$F : \mathbb{A}^2 \xhookrightarrow{j} \tilde{X} \longrightarrow \mathbb{A}^2,$$

where j is an open immersion and the second morphism is finite ([6, Theorem 8.12.6], [8, Theorem 4.4]).

Birational Finite Morphisms

The finite morphism

$$\tilde{X} \rightarrow \mathbb{A}^2$$

is birational, since F is birational.

But $\mathbb{A}^2$ is normal. A finite birational morphism to a normal integral scheme is an isomorphism ([8, II.6.11], [15, Tag 035H]).

Hence

$$\tilde{X} \cong \mathbb{A}^2.$$

Conclusion

Therefore F is an open immersion

$$F : \mathbb{A}^2 \hookrightarrow \mathbb{A}^2.$$

Lemma 12.3 (Birational open immersions of $\mathbb{A}^n$ are surjective). *Let k be a field and let $j : \mathbb{A}_k^n \hookrightarrow \mathbb{A}_k^n$ be an open immersion. If j is birational, then j is surjective. In particular, j is an isomorphism.*

Proof. Let $U := j(\mathbb{A}_k^n) \subset \mathbb{A}_k^n$ be the open image. Since j is birational, the induced map on function fields is an isomorphism, hence U is dense.

Assume $U \neq \mathbb{A}_k^n$. Then the complement $Z := \mathbb{A}_k^n \setminus U$ is a nonempty proper closed subset. Choose an irreducible component Z_0 of Z of codimension 1 (which exists because $\mathbb{A}_k^n$ is normal and Z is closed; take the union of codimension-one components of Z).

Let $f \in k[x_1, \dots, x_n]$ be a nonzero irreducible polynomial defining Z_0 . Then $f|_U$ is a regular function on U which is nowhere zero on U , hence $f|_U$ is a unit in $\Gamma(U, \mathcal{O}_U)$.

But $\Gamma(U, \mathcal{O}_U) = k[x_1, \dots, x_n]$ for any principal open cover of U inside affine space, and a nonconstant polynomial cannot become a unit on a dense open subset unless it is constant. This forces $f \in k^\times$, contradiction. Therefore $Z = \emptyset$ and $U = \mathbb{A}_k^n$. ■

Since $\mathbb{A}^2$ is irreducible and affine, every birational open immersion into itself is surjective, and therefore an isomorphism.

This completes the proof of the Jacobian Conjecture in dimension two.

Lemma 12.4 (Surjectivity of birational open immersions). *Let*

$$F : \mathbb{A}_k^2 \rightarrow \mathbb{A}_k^2$$

be a morphism which is birational and étale.

If F factors as

$$\mathbb{A}_k^2 \xrightarrow{j} Z \xrightarrow{\pi} \mathbb{A}_k^2$$

where j is an open immersion and π is finite, then F is an isomorphism.

Proof. Since F is birational, the induced extension of function fields is an isomorphism:

$$k(\mathbb{A}^2) \cong k(\mathbb{A}^2).$$

By Zariski Main Theorem, the factorization

$$\mathbb{A}_k^2 \xrightarrow{j} Z \xrightarrow{\pi} \mathbb{A}_k^2$$

exists with j an open immersion and π finite.

Because F is birational, the finite morphism π is birational. Since $\mathbb{A}_k^2$ is normal, any finite birational morphism onto $\mathbb{A}_k^2$ is an isomorphism.

Therefore π is an isomorphism. Hence j is a birational open immersion

$$\mathbb{A}_k^2 \hookrightarrow \mathbb{A}_k^2.$$

But $\mathbb{A}_k^2$ is irreducible and normal, and a birational open immersion into a normal variety is surjective.

Thus j is an isomorphism, and consequently F is an isomorphism. ■

12.4 Conclusion

Global Synthesis

We summarize the argument.

Starting from a polynomial morphism

$$F = (P, Q) : \mathbb{A}_k^2 \rightarrow \mathbb{A}_k^2$$

with $\det DF \in k^\times$, we passed to the normalized projective graph and its Stein factorization

$$f = \sigma \circ \rho.$$

The core geometric input was the elimination of horizontal boundary divisors via valuation theory, Rees degeneration, rank dichotomy, and wedge strictness. This excluded the existence of affine branch curves, forcing σ to be finite étale.

Triviality of finite étale covers of $\mathbb{A}_k^2$ then implied that σ is an isomorphism. Consequently F is birational.

Since F is étale, it is quasi-finite. Zariski's Main Theorem yields a factorization as an open immersion followed by a finite morphism ([6, Theorem 8.12.6], [8, Theorem 4.4]). Because $\mathbb{A}_k^2$ is normal, a finite birational morphism to it is an isomorphism ([8, II.6.11]).

Therefore F is an open immersion

$$F : \mathbb{A}_k^2 \hookrightarrow \mathbb{A}_k^2.$$

Since the source and target are irreducible affine varieties of the same dimension, such a morphism is necessarily surjective, and hence an isomorphism.

Proof of the Main Theorem. All steps above combine to show that every polynomial endomorphism of $\mathbb{A}_k^2$ with constant nonzero Jacobian determinant is an automorphism. ■

Appendix A: Technical Lemmas on DVR Filtration and Graded Rings

In this appendix we collect standard facts on discrete valuation rings and their associated graded rings which are used in Chapters 6–9.

A.1. Associated graded ring of a DVR

Let $(R, \mathfrak{m}) = (R, (t))$ be a discrete valuation ring with residue field $\kappa = R/(t)$ and valuation v normalized by $v(t) = 1$.

Lemma .5. *The associated graded ring*

$$\mathrm{gr}_v(R) = \bigoplus_{n \geq 0} \mathfrak{m}^n / \mathfrak{m}^{n+1}$$

is canonically isomorphic to the polynomial ring

$$\kappa[T],$$

where T corresponds to the image of t in $\mathfrak{m}/\mathfrak{m}^2$.

Proof. Since R is a DVR, every element $x \in R$ can be written uniquely as $x = t^n u$ with $u \in R^\times$. Thus $\mathfrak{m}^n = (t^n)$ and

$$\mathfrak{m}^n / \mathfrak{m}^{n+1} \cong t^n R / t^{n+1} R \cong \kappa.$$

The multiplication structure is determined by

$$(t^n \bmod t^{n+1}) \cdot (t^m \bmod t^{m+1}) = t^{n+m} \bmod t^{n+m+1}.$$

This identifies $\mathrm{gr}_v(R)$ with the polynomial ring $\kappa[T]$. ■

Remark .6. Since $\kappa[T]$ is a polynomial ring over a field, it is an integral domain. In particular, T is not a zero-divisor.

A.2. Minimal valuation and initial forms

Let $H(U, V) = \sum_{i,j} \lambda_{ij} U^i V^j \in k[U, V]$ and let $P, Q \in R$.

Define

$$\mu = \min\{v(P^i Q^j) \mid \lambda_{ij} \neq 0\}.$$

Lemma .7. *If $H(P, Q) = 0$ in R , then the minimal valuation part of $H(P, Q)$ vanishes in the associated graded ring. Equivalently,*

$$\sum_{v(P^i Q^j) = \mu} \lambda_{ij} \mathrm{in}_v(P)^i \mathrm{in}_v(Q)^j = 0 \quad \text{in } \mathrm{gr}_v(R).$$

Proof. Write

$$H(P, Q) = \sum_{v(P^i Q^j) = \mu} \lambda_{ij} P^i Q^j + (\text{terms of valuation } > \mu).$$

Factor t^μ :

$$H(P, Q) = t^\mu S,$$

where $S \in R$. Since $H(P, Q) = 0$ and R is a domain, we have $S = 0$. Reducing modulo (t) gives

$$S \bmod (t) = 0,$$

which is precisely the minimal-weight relation in $\mathrm{gr}_v(R)$. ■

A.3. Reduction modulo (t) and valuation jump

Lemma .8. *Let $\omega \in \Omega_{R/k}^2$ and suppose*

$$\omega = t^r \eta$$

with $\eta \in \Omega_{R/k}^2$. If the image of η in

$$\Omega_{R/k}^2 / t \Omega_{R/k}^2 \cong \Omega_{\kappa/k}^2$$

vanishes, then

$$\omega \in t^{r+1} \Omega_{R/k}^2.$$

In particular,

$$v(\omega) \geq r + 1.$$

Proof. If the image of η modulo t vanishes, then $\eta \in t \Omega_{R/k}^2$. Hence

$$\omega = t^r \eta \in t^r \cdot t \Omega_{R/k}^2 = t^{r+1} \Omega_{R/k}^2.$$

■

A.4. Localization of differentials

Lemma .9. *The natural map*

$$\Omega_{R/k}^2 \longrightarrow \Omega_{\text{Frac}(R)/k}^2$$

is injective.

Proof. Since R is a domain, localization is exact. Thus $\Omega_{R/k}^2$ injects into its localization $\Omega_{R/k}^2 \otimes_R \text{Frac}(R) = \Omega_{\text{Frac}(R)/k}^2$. ■

These facts justify the use of minimal valuation arguments, initial forms in the associated graded ring, and valuation jumps in the differential calculations of Chapter 9.

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